
PO11Q - Quantitative Political Analysis: From Measurement to Inference

Week 10, Exercise Solutions

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Exercises

1. In the app, set the assumed effect size to $d_{\text{test}} = 0.35$, the planned sample size to $n = 60$, the significance level to $\alpha = 0.05$, and use a two-sided test.

a. Power ≈ 0.760 .

In plain English: if the true effect size is really $d = 0.35$, and we repeatedly ran studies with $n = 60$ using a two-sided 5% significance test, we would correctly reject the null hypothesis in about 76 out of 100 studies.

```
n <- 60
d_test <- 0.35
alpha <- 0.05

df <- n - 1
lambda <- d_test * sqrt(n)
t_crit <- qt(1 - alpha / 2, df = df)

power <- pt(-t_crit, df = df, ncp = lambda) +
  (1 - pt(t_crit, df = df, ncp = lambda))

power
## [1] 0.7601439
```

This gives approximately:

```
round(power, 3)
## [1] 0.76
```

Alternatively, using the pwr package:

```
# install.packages("pwr")
library(pwr)

pwr.t.test(
  n = 60,
  d = 0.35,
  sig.level = 0.05,
  type = "one.sample",
  alternative = "two.sided"
```

```

)
##
##      One-sample t test power calculation
##
##              n = 60
##              d = 0.35
##      sig.level = 0.05
##      power = 0.7601439
##      alternative = two.sided

```

- b. The probability of a Type I Error is the significance level, $\alpha = 0.05$. So the probability of a Type I Error is 5%. In plain English: if the null hypothesis is actually true, meaning there is no real effect, we would incorrectly reject the null hypothesis in about 5 out of 100 studies. A Type I Error is therefore a false positive.
 - d. The probability of a Type II Error is $\beta = 1 - \text{Power}$. Using the power from Exercise 1a: $\beta = 1 - 0.760 \approx 0.240$. So the probability of a Type II Error is approximately **24.0%**. In plain English: if the true effect size is really $d = 0.35$, the study would fail to detect that effect in about 24 out of 100 studies. A Type II Error is therefore a false negative.
2. Reduce the planned effect size from $d_{\text{test}} = 0.35$ to $d_{\text{test}} = 0.2$.
 - a. The central t-distribution is centred at $t = 0$.
 This is because the central t-distribution represents the distribution of the test statistic under the null hypothesis, where the true effect is assumed to be zero.
 - b. Reducing the planned effect size from $d_{\text{test}} = 0.35$ to $d_{\text{test}} = 0.2$ reduces the non-centrality parameter, $\lambda = d_{\text{test}}\sqrt{n}$. Because d_{test} is smaller, the non-central t-distribution moves closer to the central t-distribution. Therefore, the overlap between the central and non-central t-distributions increases.
 - c. Power decreases because the assumed true effect is now smaller: when the true effect is smaller, the observed t-values expected under the alternative hypothesis are closer to the t-values expected under the null hypothesis. This means fewer observed t-values fall beyond the critical t-values. As a result, the probability of rejecting the null hypothesis when the alternative hypothesis is true becomes smaller.
 For example, if $n = 60$, $\alpha = 0.05$, and the test is two-sided, reducing the effect size to $d_{\text{test}} = 0.2$ gives power of approximately $\text{Power} \approx 0.332$. So the test would detect the effect only about 33.2% of the time.
 - d. To obtain approximately the same power as in Exercise 1, where $\text{Power} \approx 0.760$, with $d_{\text{test}} = 0.2$, the required sample size is approximately $n = 180$. This is much larger because smaller effects are harder to detect. The non-centrality parameter is $\lambda = d_{\text{test}}\sqrt{n}$. If d_{test} becomes smaller, n must increase to compensate. Because n is inside a square root, a substantial increase in sample size is needed to offset a reduction in effect size.

e. Using the pwr package:

```
pwr.t.test(  
  d = 0.2,  
  sig.level = 0.05,  
  power = 0.80,  
  type = "one.sample",  
  alternative = "two.sided"  
)  
##  
##      One-sample t test power calculation  
##  
##              n = 198.1508  
##              d = 0.2  
##      sig.level = 0.05  
##      power = 0.8  
##      alternative = two.sided
```

The required sample size is approximately $n = 199$.

Using base R and a search over possible sample sizes:

```
power_one_sample <- function(n, d_test, alpha = 0.05) {  
  df <- n - 1  
  lambda <- d_test * sqrt(n)  
  t_crit <- qt(1 - alpha / 2, df = df)  
  
  pt(-t_crit, df = df, ncp = lambda) +  
    (1 - pt(t_crit, df = df, ncp = lambda))  
}  
  
n_values <- 2:1000  
powers <- sapply(n_values, power_one_sample, d_test = 0.2, alpha = 0.05)  
  
n_required <- min(n_values[powers >= 0.80])  
n_required  
## [1] 199
```

3. With $d_{\text{test}} = 0.2$ and $n = 60$, reduce the significance level from 0.05 to 0.01.

- The Type II Error increases from 66.8% to 85.8%.
- For a two-sided test, the critical t-values move further away from zero. The positive critical value becomes larger, and the negative critical value becomes smaller. This means an observed t-value must be more extreme before we reject the null hypothesis. Therefore, it becomes harder to reject the null hypothesis, even when the alternative hypothesis is true. As a result, power decreases and the Type II Error rate increases.
- The central t-distribution represents the null hypothesis, and the non-central t-distribution represents the alternative hypothesis. When α is reduced, the rejection regions under the central t-distribution become smaller and move further into the tails. This reduces the probability of a Type I Error. However, because the rejection regions are now further away, a smaller proportion of the non-central t-distribution lies inside those rejection regions. Therefore, the probability of rejecting the null hypothesis when the alternative is true decreases. That means power decreases and the Type II Error rate increases.

4. We are planning a study to test whether increased news exposure increases mistrust in politicians. Based on previous research, we set $d_{\text{test}} = 0.5$ with $\alpha = 0.05$.
- For a one-sample, two-sided t-test with $d_{\text{test}} = 0.5$, $\alpha = 0.05$, and 80% power, the required sample size is approximately $n = 34$.
 - The non-centrality parameter is calculated as:

$$\lambda = d_{\text{test}}\sqrt{n}.$$

Substituting $d_{\text{test}} = 0.5$ and $n = 34$:

$$\lambda = 0.5\sqrt{34}.$$

Since $\sqrt{34} \approx 5.831$:

$$\lambda \approx 0.5 \times 5.831 = 2.916.$$

So the non-centrality parameter is approximately $\lambda \approx 2.92$.

- The non-centrality parameter λ tells us how far the alternative distribution is shifted away from the null distribution. A larger λ means the central and non-central t-distributions are further apart, there is less overlap between them, and the test has higher power.
- No, if the sample size doubled, λ would not double.

This is because $\lambda = d_{\text{test}}\sqrt{n}$. The sample size enters the formula through its square root. If the sample size is doubled from n to $2n$, then:

$$\lambda_{\text{new}} = d_{\text{test}}\sqrt{2n} = \sqrt{2}d_{\text{test}}\sqrt{n} = \sqrt{2}\lambda.$$

Since $\sqrt{2} \approx 1.414$, doubling the sample size would multiply λ by about 1.414, not by 2.

5. A lecturer in the Statistics department wants to conduct a power analysis on the change of exam marks, following the introduction of a new university policy. Her colleague from the History department has already conducted such a study and has found that the effect size of the policy change in his department is $d_{\text{obs}} = 0.3$ in a study designed to achieve 80% power. Given the nature of the taught material, the Statistics lecturer expects larger variation amongst the students in her department, however.
- The Statistics lecturer should not simply set $d_{\text{test}} = 0.3$. She should choose a smaller test effect size that reflects the expected variability in Statistics marks.
- Cohen's d is calculated as the mean difference divided by the standard deviation:

$$d = \frac{\text{mean difference}}{\text{standard deviation}}.$$

If marks in Statistics have greater variation, their standard deviation will be larger. For the same mean change in marks, this leads to a smaller standardised effect size. Therefore, the effect observed in History may not generalise to Statistics, and using $d_{\text{test}} = 0.3$ may be too optimistic.

Moreover, $d_{\text{obs}} = 0.3$ is an estimate from only one previous study and is thus liable to sampling error. The lecturer should therefore select a smaller value of d_{test} , informed by relevant prior data, a pilot study, or by combining what would constitute a substantively meaningful raw change in marks with an estimate of the standard deviation.

- b. No. Increasing the sample size increases power, but an arbitrary increase does not necessarily ensure that the Statistics study will have at least as much *actual* power as the History study.

For a one-sample test, power is related to the non-centrality parameter

$$\lambda = d\sqrt{n}.$$

A larger sample size increases λ and therefore increases power. However, as explained in part (a), the greater variation in Statistics marks means that the relevant effect size is likely to be smaller than 0.3. A smaller effect size reduces λ and therefore reduces power. The larger sample size and the smaller effect size therefore pull in opposite directions.

Our issue here is that an increase in n is being considered whilst retaining a d_{test} value which is likely to be too high for the Statistics department. If the lecturer calculates power using $d_{\text{test}} = 0.3$, the larger sample will produce a higher *calculated* power than in the History study. However, this calculation may overstate the Statistics study's *actual* power because it is based on an effect size that is likely to be too large.

- c. As a result, the lecturer needs to make several adjustments:

- She should choose a smaller, defensible value of d_{test} that reflects the expected greater variability in Statistics marks, rather than simply using the observed History effect of $d_{\text{obs}} = 0.3$. This value should be informed by relevant prior data, a pilot study, or by combining what would constitute a substantively meaningful raw change in marks with an estimate of the standard deviation.
- She can use the 80% power achieved by the History study as her target as this is the commonly accepted threshold.
- Holding the design and significance level fixed, she should calculate the sample size required to attain at least 80% power using the revised value of d_{test} . She should not simply choose an arbitrary sample size larger than the History sample.
- Because the appropriate effect size is uncertain, she should repeat the power analysis for a range of plausible, smaller effect sizes. She can then choose the corresponding sample sizes that provide the desired 80% power.

Importantly, no power analysis can guarantee a particular level of actual power when the true effect size is unknown. But these steps provide a defensible plan for designing a study intended to have at least as much power as the History study.

6. Suppose the Statistics lecturer from the previous exercise knows that the long-established average of marks in her department is 72. She has now marked the first 20 students and found that the average mark is 75. She wants to test whether this is a significant increase in marks.

- a. The null hypothesis is that the average mark has not increased, formally $H_0 : \mu \leq 72$, because the lecturer is specifically interested in whether marks have increased.
- b. The alternative hypothesis is that the average mark has increased, formally $H_1 : \mu > 72$.
This is a one-sided alternative hypothesis because the lecturer is specifically testing for an increase in marks.
- c. The t-statistic is calculated as:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

Here, $\bar{x} = 75$, $\mu_0 = 72$, $s = 10$, and $n = 20$. Therefore:

$$t = \frac{75 - 72}{10/\sqrt{20}}.$$

The numerator is $75 - 72 = 3$, and the standard error is:

$$\frac{10}{\sqrt{20}} \approx 2.236.$$

Therefore:

$$t \approx \frac{3}{2.236} \approx 1.342.$$

So the observed t-statistic is approximately $t \approx 1.34$.

- d. The observed effect size is:

$$d_{\text{obs}} = \frac{\bar{x} - \mu_0}{s}.$$

Substituting the values:

$$d_{\text{obs}} = \frac{75 - 72}{10} = \frac{3}{10} = 0.3.$$

$d_{\text{obs}} = 0.3$ would usually be interpreted as a small to medium effect.

- e. We can show that $t = d_{\text{obs}}\sqrt{n}$ using these numbers:

$$d_{\text{obs}}\sqrt{n} = 0.3\sqrt{20} \approx 0.3 \times 4.472 = 1.342.$$

This is the same value as the t-statistic calculated in Exercise 6c.